Chapter 9. Experiments with Random Factors

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1 Introduction: Fixed vs Random Factors

- **Fixed factor**: a specific set of levels is chosen; inference is confined to those levels. Quantitative factors are usually fixed (levels chosen to span a region for a response curve).
- **Random factor**: levels are chosen at random from a larger population of potential levels; inference is about the *entire population* of levels. Industrial applications include measurement-system (gauge R&R) studies.

2 The One-Way Random Effects Model

2.1 Example: looms

Fabric is woven on hundreds of looms; do looms differ in strength? Four looms chosen *at random*, four determinations each ($a = 4$, $n = 4$): a random factor.

Loom	Observations				$y_{i\cdot}$
1	98	97	99	96	390
2	91	90	93	92	366
3	96	95	97	95	383
4	95	96	99	98	388
					1527

2.2 Model and covariance structure

Definition 9.1

$$y_{ij} = \mu + \tau_i + \varepsilon_{ij}, \qquad i = 1, \dots, a;\ j = 1, \dots, n,$$

$\tau_i \overset{\text{iid}}{\sim} N(0, \sigma_\tau^2)$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, $\tau_i \perp\!\!\!\perp \varepsilon_{ij}$. σ_τ^2 and σ^2 are **variance components**.

Derivation

$$\begin{aligned}
\mathrm{E}(y_{ij}) &= \mu, \qquad \mathrm{Var}(y_{ij}) = \mathrm{Var}(\tau_i) + \mathrm{Var}(\varepsilon_{ij}) = \sigma_\tau^2 + \sigma^2, \\
\mathrm{Cov}(y_{ij}, y_{ij'}) &= \mathrm{Cov}(\tau_i + \varepsilon_{ij}, \tau_i + \varepsilon_{ij'}) = \mathrm{E}\big[(\tau_i + \varepsilon_{ij})(\tau_i + \varepsilon_{ij'})\big] = \mathrm{E}(\tau_i^2) = \sigma_\tau^2 \quad (j \neq j'), \\
\mathrm{Cov}(y_{ij}, y_{i'j'}) &= \mathrm{E}\big[(\tau_i + \varepsilon_{ij})(\tau_{i'} + \varepsilon_{i'j'})\big] = 0 \quad (i \neq i').
\end{aligned}$$

So with $\mathbf{y} = (y_{11}, \dots, y_{1n}, \dots, y_{a1}, \dots, y_{an})^T$,

$$\boldsymbol{\Sigma} = \mathrm{Cov}(\mathbf{y}) = \begin{pmatrix} \sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n & & \\ & \ddots & \\ & & \sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n \end{pmatrix} = \mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n),$$

a block-diagonal matrix: observations from the same loom are correlated (intraclass correlation $\rho = \sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$), different looms are independent. Under $H_0 : \sigma_\tau^2 = 0$, $\mathrm{Cov}(\mathbf{y}) = \sigma^2\mathbf{I}$.

2.3 Hypotheses

The individual treatments are not of specific interest; we care about the population:

$H_0 : \sigma_\tau^2 = 0$ (all treatments identical) vs $H_1 : \sigma_\tau^2 > 0$ (variability exists between treatments).

2.4 ANOVA and expected mean squares

The partition $SS_T = SS_{\text{Treatments}} + SS_E$ with df $an - 1 = (a-1) + a(n-1)$ is purely algebraic and still holds.

Theorem 9.2

$$\mathrm{E}(MS_{\text{Treatments}}) = \sigma^2 + n\sigma_\tau^2, \qquad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\varepsilon}_{i\cdot}$ and $\bar{y}_{\cdot\cdot} = \mu + \bar{\tau} + \bar{\varepsilon}_{\cdot\cdot}$ (now $\bar{\tau} = \frac{1}{a}\sum \tau_i$ is random, not 0). Thus

$$\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot} = (\tau_i - \bar{\tau}) + (\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}),$$

and the two brackets are independent with mean zero, so the cross term has expectation zero:

$$\mathrm{E}(MS_{\text{Treatments}}) = \mathrm{E}\Big[\frac{n\sum_i(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2}{a-1}\Big] = n\underbrace{\mathrm{E}\Big[\frac{\sum_i(\tau_i - \bar{\tau})^2}{a-1}\Big]}_{=\mathrm{Var}(\tau_i)=\sigma_\tau^2} + n\underbrace{\mathrm{E}\Big[\frac{\sum_i(\bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot})^2}{a-1}\Big]}_{=\mathrm{Var}(\bar{\varepsilon}_{i\cdot})=\sigma^2/n} = n\sigma_\tau^2 + \sigma^2.$$

(A sample variance of iid variables is unbiased for their variance.) For the error, $y_{ij} - \bar{y}_{i\cdot} = \varepsilon_{ij} - \bar{\varepsilon}_{i\cdot}$ (τ_i cancels), so

$$\mathrm{E}(MS_E) = \mathrm{E}\Big[\frac{\sum\sum(y_{ij} - \bar{y}_{i\cdot})^2}{a(n-1)}\Big] = \frac{1}{a}\sum_{i=1}^{a}\mathrm{E}\Big[\frac{\sum_j(\varepsilon_{ij} - \bar{\varepsilon}_{i\cdot})^2}{n-1}\Big] = \sigma^2.$$

■

Test: $F_0 = MS_{\text{Treatments}}/MS_E \overset{H_0}{\sim} F_{a-1,N-a}$; reject $H_0 : \sigma_\tau^2 = 0$ if $F_0 > F_{\alpha,a-1,N-a}$. (Same computation as the fixed case; only the hypothesis and interpretation differ.)

2.5 Quadratic-form treatment

Theorem 9.3

Let $\mathbf{A} = \frac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n - \frac{1}{N}\mathbf{J}_N$. Then (a) $SS_{\text{Treatments}} = \mathbf{y}^T\mathbf{A}\mathbf{y}$;
(b) $\mathbf{A}\boldsymbol{\Sigma} = (\sigma^2 + n\sigma_\tau^2)\mathbf{A}$, so $\mathbf{A}\boldsymbol{\Sigma}/(\sigma^2 + n\sigma_\tau^2) = \mathbf{A}$ is idempotent and $SS_{\text{Treatments}}/(\sigma^2 + n\sigma_\tau^2) \sim \chi^2_{a-1}$;
(c) $\mathrm{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = (a-1)(\sigma^2 + n\sigma_\tau^2)$.

Proof [Supplement]. (a) $\frac{1}{n}(\mathbf{I}_a \otimes \mathbf{J}_n)\mathbf{y}$ replaces each y_{ij} by $\bar{y}_{i\cdot}$, and $\frac{1}{N}\mathbf{J}_N\mathbf{y} = \bar{y}_{\cdot\cdot}\mathbf{1}$. So $\mathbf{A}\mathbf{y}$ is the vector of $\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot}$. $\mathbf{A}$ is symmetric and idempotent (it is $\mathbf{H} - \frac{1}{N}\mathbf{J}_N$ of Ch. 3), hence $\mathbf{y}^T\mathbf{A}\mathbf{y} = (\mathbf{A}\mathbf{y})^T(\mathbf{A}\mathbf{y}) = \sum_i\sum_j(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2$.

(b) Using $(\mathbf{P}\otimes\mathbf{Q})(\mathbf{R}\otimes\mathbf{S}) = \mathbf{PR}\otimes\mathbf{QS}$ and $\mathbf{J}_n\mathbf{J}_n = n\mathbf{J}_n$:

$$\tfrac{1}{n}(\mathbf{I}_a \otimes \mathbf{J}_n)(\mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n)) = \mathbf{I}_a \otimes \tfrac{1}{n}(\sigma^2\mathbf{J}_n + n\sigma_\tau^2\mathbf{J}_n) = (\sigma^2 + n\sigma_\tau^2)\,\tfrac{1}{n}\mathbf{I}_a \otimes \mathbf{J}_n,$$
$$\tfrac{1}{N}(\mathbf{J}_a \otimes \mathbf{J}_n)(\mathbf{I}_a \otimes (\sigma^2\mathbf{I}_n + \sigma_\tau^2\mathbf{J}_n)) = \tfrac{1}{N}\mathbf{J}_a \otimes (\sigma^2 + n\sigma_\tau^2)\mathbf{J}_n = (\sigma^2 + n\sigma_\tau^2)\tfrac{1}{N}\mathbf{J}_N.$$

Subtracting, $\mathbf{A\Sigma} = (\sigma^2 + n\sigma_\tau^2)\mathbf{A}$. For $\mathbf{y} \sim N(\mu\mathbf{1}, \mathbf{\Sigma})$, $\mathbf{y}^T\mathbf{B}\mathbf{y} \sim \chi^2_{\text{rank}\mathbf{B}}(\cdot)$ when $\mathbf{B\Sigma}$ is idempotent; with $\mathbf{B} = \mathbf{A}/(\sigma^2 + n\sigma_\tau^2)$, $\mathbf{B\Sigma} = \mathbf{A}$ is idempotent of rank $\text{tr}\mathbf{A} = a - 1$, and the noncentrality $\mu^2\mathbf{1}^T\mathbf{B}\mathbf{1} = 0$ since $\mathbf{A}\mathbf{1} = \mathbf{0}$.

(c) $\text{E}(\mathbf{y}^T\mathbf{A}\mathbf{y}) = \text{tr}(\mathbf{A\Sigma}) + \mu^2\mathbf{1}^T\mathbf{A}\mathbf{1} = (\sigma^2 + n\sigma_\tau^2)\text{tr}(\mathbf{A}) + 0 = (a-1)(\sigma^2 + n\sigma_\tau^2)$. ■

2.6 Estimating variance components

ANOVA (moment) estimators, from equating MS to EMS:

$$\hat{\sigma}^2 = MS_E, \qquad \hat{\sigma}_\tau^2 = \frac{MS_{\text{Treatments}} - MS_E}{n}, \qquad \widehat{\text{Var}}(y_{ij}) = \hat{\sigma}_y^2 = \hat{\sigma}_\tau^2 + \hat{\sigma}^2.$$

Unbiased: $\text{E}(\hat{\sigma}_\tau^2) = \frac{(\sigma^2 + n\sigma_\tau^2) - \sigma^2}{n} = \sigma_\tau^2$. Also $\hat{\sigma}_\tau^2 = \widehat{\text{Cov}}(y_{ij}, y_{ij'})$ and $\widehat{\text{Cov}}(\mathbf{y})$ has $\hat{\sigma}^2 + \hat{\sigma}_\tau^2$ on the diagonal blocks and $\hat{\sigma}_\tau^2$ off-diagonal within blocks.

Potential problems:

(1) **negative estimates** when $MS_{\text{Treatments}} < MS_E$;

(2) they are moment estimators and do not have the best statistical properties (REML/ML via `lme` are alternatives).

2.7 Confidence intervals

For σ^2. Under normality $\frac{(N-a)MS_E}{\sigma^2} \sim \chi^2_{N-a}$, so

$$P\Big(\chi^2_{1-\alpha/2,N-a} \le \frac{(N-a)MS_E}{\sigma^2} \le \chi^2_{\alpha/2,N-a}\Big) = 1 - \alpha \;\Rightarrow\; \frac{(N-a)MS_E}{\chi^2_{\alpha/2,N-a}} \le \sigma^2 \le \frac{(N-a)MS_E}{\chi^2_{1-\alpha/2,N-a}}.$$

For σ_τ^2: no exact interval exists.

For the intraclass correlation $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$. Since $\frac{(a-1)MS_{\text{Treatments}}}{\sigma^2 + n\sigma_\tau^2} \sim \chi^2_{a-1}$ and is independent of $\frac{(N-a)MS_E}{\sigma^2} \sim \chi^2_{N-a}$,

$$\mathcal{A} = \frac{MS_{\text{Treatments}}/(n\sigma_\tau^2 + \sigma^2)}{MS_E/\sigma^2} \sim F_{a-1,N-a} \quad \Rightarrow \quad P(F_{1-\alpha/2,a-1,N-a} \le \mathcal{A} \le F_{\alpha/2,a-1,N-a}) = 1 - \alpha.$$

Inverting the pivot [Supplement]. Let $\theta = \sigma_\tau^2/\sigma^2$ and $F^* = MS_{\text{Treatments}}/MS_E$. Then $\mathcal{A} = F^*/(1 + n\theta)$ and

$$F_{1-\alpha/2} \le \frac{F^*}{1+n\theta} \le F_{\alpha/2} \iff \frac{1}{n}\Big(\frac{F^*}{F_{\alpha/2}} - 1\Big) \le \theta \le \frac{1}{n}\Big(\frac{F^*}{F_{1-\alpha/2}} - 1\Big).$$

Call these bounds L and U. Since $\frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2} = \frac{\theta}{1+\theta}$ is increasing in θ,

$$\frac{L}{1+L} \le \frac{\sigma_\tau^2}{\sigma_\tau^2 + \sigma^2} \le \frac{U}{1+U},$$

$$L = \frac{1}{n}\Big(\frac{MS_{\text{Treatments}}}{MS_E}\frac{1}{F_{\alpha/2,a-1,N-a}} - 1\Big), \quad U = \frac{1}{n}\Big(\frac{MS_{\text{Treatments}}}{MS_E}\frac{1}{F_{1-\alpha/2,a-1,N-a}} - 1\Big).$$

■

Example 9.4: Looms, line by line

$\sum y^2 = 145845$, $y_{\cdot\cdot}^2/N = 1527^2/16 = 145733.06$.

$$SS_T = 145845 - 145733.06 = 111.94,$$
$$SS_{\text{Looms}} = \frac{390^2 + 366^2 + 383^2 + 388^2}{4} - 145733.06 = \frac{583289}{4} - 145733.06 = 89.19,$$
$$SS_E = 111.94 - 89.19 = 22.75.$$

Source	SS	DF	MS	F_0
Looms	89.19	3	29.73	15.68
Error	22.75	12	1.90	
Total	111.94	15		

$F_0 = 15.68 > F_{0.05,3,12} = 3.49$ ($P = 0.00019$): reject $H_0 : \sigma_\tau^2 = 0$.

$$\hat\sigma^2 = 1.90, \qquad \hat\sigma_\tau^2 = \frac{29.73 - 1.90}{4} = 6.96, \qquad \hat\sigma_y^2 = 6.96 + 1.90 = 8.86$$

(`lme`: 6.958 and 1.896). Most of the variability ($6.96/8.86 = 78.6\%$) is due to looms.
95% CI for σ^2: $\chi^2_{0.025,12} = 23.34$, $\chi^2_{0.975,12} = 4.40$: $[\frac{12(1.896)}{23.34}, \frac{12(1.896)}{4.40}] = [0.975,\ 5.17]$.
95% CI for $\sigma_\tau^2/(\sigma_\tau^2 + \sigma^2)$: $F_{0.025,3,12} = 4.47$, $F_{0.975,3,12} = 0.0698$, $F^* = 15.68$:

$$L = \tfrac{1}{4}(\tfrac{15.68}{4.47} - 1) = 0.626, \quad U = \tfrac{1}{4}(\tfrac{15.68}{0.0698} - 1) = 55.95, \quad \left[\tfrac{0.626}{1.626},\ \tfrac{55.95}{56.95}\right] = [0.385,\ 0.982].$$

3 Two-Factor Factorial with Random Factors

Motivation. Randomly sampled levels contribute a variance; expected mean squares decide the F-ratio.

Definition 9.5

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk},$$

$\tau_i \sim N(0, \sigma_\tau^2)$, $\beta_j \sim N(0, \sigma_\beta^2)$, $(\tau\beta)_{ij} \sim N(0, \sigma_{\tau\beta}^2)$, $\varepsilon_{ijk} \sim N(0, \sigma^2)$, all independent.
$\text{Var}(y_{ijk}) = \sigma_\tau^2 + \sigma_\beta^2 + \sigma_{\tau\beta}^2 + \sigma^2$.

Hypotheses: $H_0 : \sigma_\tau^2 = 0$, $H_0 : \sigma_\beta^2 = 0$, $H_0 : \sigma_{\tau\beta}^2 = 0$. The ANOVA computations are unchanged; the tests change.

Theorem 9.6

$$\text{E}(MS_A) = \sigma^2 + n\sigma_{\tau\beta}^2 + bn\sigma_\tau^2, \quad \text{E}(MS_B) = \sigma^2 + n\sigma_{\tau\beta}^2 + an\sigma_\beta^2,$$
$$\text{E}(MS_{AB}) = \sigma^2 + n\sigma_{\tau\beta}^2, \quad \text{E}(MS_E) = \sigma^2.$$

$\text{E}(MS_A)$ **[Supplement].**

$$\bar y_{i\cdot\cdot} = \mu + \tau_i + \bar\beta + \overline{(\tau\beta)}_{i\cdot} + \bar\varepsilon_{i\cdot\cdot}, \qquad \bar y_{\cdots} = \mu + \bar\tau + \bar\beta + \overline{(\tau\beta)}_{\cdot\cdot} + \bar\varepsilon_{\cdots},$$
$$SS_A = bn \sum_i \left[(\tau_i - \bar\tau) + (\overline{(\tau\beta)}_{i\cdot} - \overline{(\tau\beta)}_{\cdot\cdot}) + (\bar\varepsilon_{i\cdot\cdot} - \bar\varepsilon_{\cdots})\right]^2.$$

$\bar\beta$ cancels. The three brackets are independent with mean 0, so cross terms vanish in expectation, and each is a sample-variance-type sum of iid terms:

$$\mathrm{E}(MS_A) = bn\big[\mathrm{Var}(\tau_i) + \mathrm{Var}(\overline{(\tau\beta)}_{i\cdot}) + \mathrm{Var}(\bar\varepsilon_{i\cdot\cdot})\big] = bn\Big(\sigma_\tau^2 + \frac{\sigma_{\tau\beta}^2}{b} + \frac{\sigma^2}{bn}\Big) = bn\sigma_\tau^2 + n\sigma_{\tau\beta}^2 + \sigma^2.$$

By symmetry $\mathrm{E}(MS_B) = an\sigma_\beta^2 + n\sigma_{\tau\beta}^2 + \sigma^2$. For MS_{AB}: $\bar y_{ij\cdot} - \bar y_{i\cdot\cdot} - \bar y_{\cdot j\cdot} + \bar y_{\cdot\cdot\cdot}$ removes τ_i, β_j and leaves the RCBD-type residual of the table $\{(\tau\beta)_{ij} + \bar\varepsilon_{ij\cdot}\}$ of iid variables with variance $\sigma_{\tau\beta}^2 + \sigma^2/n$; its expected sum of squares is $(a-1)(b-1)(\sigma_{\tau\beta}^2 + \sigma^2/n)$, hence $\mathrm{E}(MS_{AB}) = n\sigma_{\tau\beta}^2 + \sigma^2$. ■

Tests (choose the denominator whose EMS equals the numerator's under H_0):

$$H_0 : \sigma_\tau^2 = 0 : \ F_0 = \frac{MS_A}{MS_{AB}}, \qquad H_0 : \sigma_\beta^2 = 0 : \ F_0 = \frac{MS_B}{MS_{AB}}, \qquad H_0 : \sigma_{\tau\beta}^2 = 0 : \ F_0 = \frac{MS_{AB}}{MS_E}.$$

Estimators:

$$\hat\sigma_\tau^2 = \frac{MS_A - MS_{AB}}{bn}, \quad \hat\sigma_\beta^2 = \frac{MS_B - MS_{AB}}{an}, \quad \hat\sigma_{\tau\beta}^2 = \frac{MS_{AB} - MS_E}{n}, \quad \hat\sigma^2 = MS_E.$$

Example 9.7: Gauge capability study

20 parts (random) × 3 operators (random), $n = 2$ measurements each.

Source	SS	DF	MS	F_0	denominator
Parts	1185.43	19	62.39	$62.39/0.71 = 87.9$	MS_{AB}
Operators	2.62	2	1.31	$1.31/0.71 = 1.84$	MS_{AB}
P×O	27.05	38	0.71	$0.71/0.99 = 0.72$	MS_E
Error	59.50	60	0.99		

$$\hat\sigma_\tau^2 = \frac{62.39 - 0.71}{3\cdot 2} = 10.28, \quad \hat\sigma_\beta^2 = \frac{1.31 - 0.71}{20\cdot 2} = 0.015,$$
$$\hat\sigma_{\tau\beta}^2 = \frac{0.71 - 0.99}{2} = -0.14, \quad \hat\sigma^2 = 0.99.$$

The negative $\hat\sigma_{\tau\beta}^2$ is consistent with a zero interaction component; one may refit the reduced model $y = \mu + \tau_i + \beta_j + \varepsilon$. Gauge variability $\hat\sigma_{\text{gauge}}^2 = \hat\sigma^2 + \hat\sigma_\beta^2$ (repeatability + reproducibility) is small relative to part variability.

4 Two-Factor Mixed Models

Motivation. In a mixed model the fixed factor is tested against the interaction mean square.

4.1 Restricted mixed model

Definition 9.8: A fixed, B random (restricted model)

$$y_{ijk} = \mu + \tau_i + \beta_j + (\tau\beta)_{ij} + \varepsilon_{ijk},$$

$\sum_i \tau_i = 0$; $\beta_j \overset{\text{iid}}{\sim} N(0, \sigma_\beta^2)$; $(\tau\beta)_{ij} \sim N\big(0, \frac{a-1}{a}\sigma_{\tau\beta}^2\big)$ with the restriction $\sum_{i=1}^{a}(\tau\beta)_{ij} = (\tau\beta)_{\cdot j} = 0$ for each j; $\varepsilon_{ijk} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

Interaction terms in the same column are therefore *not* independent:

Theorem 9.9

$\mathrm{Cov}[(\tau\beta)_{ij},(\tau\beta)_{i'j}] = -\frac{1}{a}\sigma^2_{\tau\beta}$ $(i \neq i')$, and $\mathrm{Cov}[(\tau\beta)_{ij},(\tau\beta)_{ij'}] = 0$ $(j \neq j')$.

Proof [Supplement]. Let $x = \mathrm{Cov}[(\tau\beta)_{ij},(\tau\beta)_{i'j}]$ (same for all $i \neq i'$ by symmetry). Since $\sum_i (\tau\beta)_{ij} = 0$,

$$0 = \mathrm{Var}\Big(\sum_{i=1}^{a}(\tau\beta)_{ij}\Big) = \sum_{i=1}^{a}\mathrm{Var}((\tau\beta)_{ij}) + 2\binom{a}{2}x = a\cdot\frac{a-1}{a}\sigma^2_{\tau\beta} + a(a-1)x \;\Rightarrow\; x = -\frac{\sigma^2_{\tau\beta}}{a}.$$

Different columns j correspond to independently drawn random levels, so their interactions are uncorrelated. ■

Theorem 9.10: EMS, restricted model

$$\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\tau\beta} + \frac{bn\sum_i \tau_i^2}{a-1}, \quad \mathrm{E}(MS_B) = \sigma^2 + an\sigma^2_{\beta},$$
$$\mathrm{E}(MS_{AB}) = \sigma^2 + n\sigma^2_{\tau\beta}, \quad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. MS_A. $\bar{y}_{i\cdot\cdot} = \mu + \tau_i + \bar{\beta} + \overline{(\tau\beta)}_{i\cdot} + \bar{\varepsilon}_{i\cdot\cdot}$; $\bar{y}_{\cdots} = \mu + \bar{\tau} + \bar{\beta} + \overline{(\tau\beta)}_{\cdot\cdot} + \bar{\varepsilon}_{\cdots}$ with $\bar{\tau} = 0$ and $\overline{(\tau\beta)}_{\cdot\cdot} = 0$ (restriction). So $\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots} = \tau_i + \overline{(\tau\beta)}_{i\cdot} + (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})$ and

$$\mathrm{E}(MS_A) = bn\frac{\sum\tau_i^2}{a-1} + bn\underbrace{\mathrm{E}\Big(\frac{\sum_i \overline{(\tau\beta)}_{i\cdot}^2}{a-1}\Big)}_{(*)} + bn\underbrace{\mathrm{E}\Big(\frac{\sum_i(\bar{\varepsilon}_{i\cdot\cdot}-\bar{\varepsilon}_{\cdots})^2}{a-1}\Big)}_{=\sigma^2/(bn)}.$$

$(*)$: $\overline{(\tau\beta)}_{i\cdot} = \frac{1}{b}\sum_j(\tau\beta)_{ij}$ is an average of b independent terms (different j), so

$$\mathrm{E}\Big(\sum_{i=1}^{a}\overline{(\tau\beta)}_{i\cdot}^2\Big) = \sum_i \mathrm{Var}(\overline{(\tau\beta)}_{i\cdot}) = \sum_{i=1}^{a}\frac{\mathrm{Var}((\tau\beta)_{ij})}{b} = \frac{a}{b}\cdot\frac{a-1}{a}\sigma^2_{\tau\beta} = \frac{a-1}{b}\sigma^2_{\tau\beta},$$

so $bn(*) = bn\cdot\frac{\sigma^2_{\tau\beta}}{b} = n\sigma^2_{\tau\beta}$. Hence $\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\tau\beta} + \frac{bn\sum\tau_i^2}{a-1}$.

MS_B. $\bar{y}_{\cdot j\cdot} = \mu + \bar{\tau} + \beta_j + \overline{(\tau\beta)}_{\cdot j} + \bar{\varepsilon}_{\cdot j\cdot} = \mu + \beta_j + \bar{\varepsilon}_{\cdot j\cdot}$ because $\bar{\tau} = 0$ and $\overline{(\tau\beta)}_{\cdot j} = 0$ by the restriction. Thus $\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots} = (\beta_j - \bar{\beta}) + (\bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}_{\cdots})$ and

$$\mathrm{E}(MS_B) = an\,\mathrm{E}\Big(\frac{\sum_j(\beta_j-\bar{\beta})^2}{b-1}\Big) + an\,\mathrm{E}\Big(\frac{\sum_j(\bar{\varepsilon}_{\cdot j\cdot}-\bar{\varepsilon}_{\cdots})^2}{b-1}\Big) = an\sigma^2_{\beta} + an\frac{\sigma^2}{an} = \sigma^2 + an\sigma^2_{\beta}.$$

The interaction does *not* appear in $\mathrm{E}(MS_B)$ in the restricted model.

MS_{AB}. $\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdots} = ((\tau\beta)_{ij} - \overline{(\tau\beta)}_{i\cdot}) + (\varepsilon\text{-part})$. For fixed i, $\{(\tau\beta)_{ij}\}_{j=1}^{b}$ are iid with variance $\frac{a-1}{a}\sigma^2_{\tau\beta}$, so $\mathrm{E}\sum_j((\tau\beta)_{ij} - \overline{(\tau\beta)}_{i\cdot})^2 = (b-1)\frac{a-1}{a}\sigma^2_{\tau\beta}$; summing over i: $(a-1)(b-1)\sigma^2_{\tau\beta}$. With the error part contributing $(a-1)(b-1)\sigma^2/n$, $\mathrm{E}(MS_{AB}) = n\sigma^2_{\tau\beta} + \sigma^2$. ■

Tests: $H_0 : \tau_i = 0$: $F_0 = MS_A/MS_{AB}$; $H_0 : \sigma^2_{\beta} = 0$: $F_0 = MS_B/MS_E$; $H_0 : \sigma^2_{\tau\beta} = 0$: $F_0 = MS_{AB}/MS_E$.
Estimators: $\hat{\sigma}^2_{\beta} = \frac{MS_B - MS_E}{an}$, $\hat{\sigma}^2_{\tau\beta} = \frac{MS_{AB}-MS_E}{n}$, $\hat{\sigma}^2 = MS_E$, $\hat{\tau}_i = \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots}$.

4.2 Unrestricted mixed model

Definition 9.11

$$y_{ijk} = \mu + \alpha_i + \gamma_j + (\alpha\gamma)_{ij} + \varepsilon_{ijk},$$

$$\sum_i \alpha_i = 0, \quad \gamma_j \sim N(0, \sigma^2_\gamma), \quad (\alpha\gamma)_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2_{\alpha\gamma}), \quad \varepsilon_{ijk} \sim N(0, \sigma^2),$$

with *no* sum-to-zero restriction on the interactions (all mutually independent).

Theorem 9.12: EMS, unrestricted model

$$\mathrm{E}(MS_A) = \sigma^2 + n\sigma^2_{\alpha\gamma} + \frac{bn\sum \alpha_i^2}{a-1}, \quad \mathrm{E}(MS_B) = \sigma^2 + n\sigma^2_{\alpha\gamma} + an\sigma^2_\gamma,$$

$$\mathrm{E}(MS_{AB}) = \sigma^2 + n\sigma^2_{\alpha\gamma}, \quad \mathrm{E}(MS_E) = \sigma^2.$$

Proof [Supplement]. $\bar{y}_{i\cdot\cdot} - \bar{y}_{\cdots} = \alpha_i + [\overline{(\alpha\gamma)}_{i\cdot} - \overline{(\alpha\gamma)}_{\cdot\cdot}] + (\bar{\varepsilon}_{i\cdot\cdot} - \bar{\varepsilon}_{\cdots})$ ($\bar{\gamma}$ cancels, $\bar{\alpha} = 0$). The $\overline{(\alpha\gamma)}_{i\cdot}$ are iid with variance $\sigma^2_{\alpha\gamma}/b$, so

$$\mathrm{E}(MS_A) = \frac{bn\sum \alpha_i^2}{a-1} + bn \cdot \frac{\sigma^2_{\alpha\gamma}}{b} + bn \cdot \frac{\sigma^2}{bn} = \frac{bn\sum \alpha_i^2}{a-1} + n\sigma^2_{\alpha\gamma} + \sigma^2.$$

$\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdots} = (\gamma_j - \bar{\gamma}) + [\overline{(\alpha\gamma)}_{\cdot j} - \overline{(\alpha\gamma)}_{\cdot\cdot}] + (\bar{\varepsilon}_{\cdot j\cdot} - \bar{\varepsilon}_{\cdots})$ with $\mathrm{Var}(\overline{(\alpha\gamma)}_{\cdot j}) = \sigma^2_{\alpha\gamma}/a$, so

$$\mathrm{E}(MS_B) = an\sigma^2_\gamma + an \cdot \frac{\sigma^2_{\alpha\gamma}}{a} + an \cdot \frac{\sigma^2}{an} = an\sigma^2_\gamma + n\sigma^2_{\alpha\gamma} + \sigma^2.$$

■

Consequently, in the unrestricted model $H_0 : \sigma^2_\gamma = 0$ is tested by $F_0 = MS_B/MS_{AB}$ and $\hat{\sigma}^2_\gamma = \frac{MS_B - MS_{AB}}{an}$. (Restricted vs unrestricted differ only in the test/estimator for the random main effect.)

5 Rules for Expected Mean Squares

1. The error term $\varepsilon_{ijk\ldots m}$ is written $\varepsilon_{(ijk\ldots)m}$, where m is the replication subscript.
2. For each term, classify subscripts: **live** (present, not in parentheses), **dead** (present, in parentheses), **absent** (in the model but not in that term). E.g. $(\tau\beta)_{ij}$: i, j live, k absent; $\varepsilon_{(ij)k}$: k live, i, j dead.
3. DF of a term $= \prod$(#levels of dead subscripts) $\times \prod$(#levels -1 of live subscripts). E.g. $(\tau\beta)_{ij}$: $(a-1)(b-1)$; $\varepsilon_{(ij)k}$: $ab(n-1)$.
4. If an interaction contains at least one random effect, the entire interaction is random.
5. Variance components: $\sigma^2_\tau, \sigma^2_\beta, \ldots$; fixed factors: $\sum \tau_i^2/(a-1)$, $\sum\sum(\tau\beta)_{ij}^2/((a-1)(b-1))$, …
6. Replicates are always random.

Steps (restricted model):

S1. Write the terms of the model as row labels; column headings are the subscripts, with F/R and the number of levels above them.

S2. In each row, write 1 in a column if the column's subscript is *dead* in that row's term.

S3. In each row, if the column subscript is *live* in the term, write 0 if the column factor is fixed (F) and 1 if random (R).

S4. Fill the remaining cells (absent subscripts) with the number of levels shown above the column.

S5. For the EMS of a term, cover the columns headed by the term's live subscripts.

S6. Consider every row that contains at least all the subscripts of the term; multiply the remaining numbers in that row and attach the row's fixed factor or variance component. Sum these.

Example 9.13: Two-factor mixed model via the algorithm (A fixed, B random)

	F	R	R	
	a	b	n	
Term	i	j	k	EMS
τ_i	0	b	n	$\sigma^2 + n\sigma^2_{\tau\beta} + bn\sum \tau_i^2/(a-1)$
β_j	a	1	n	$\sigma^2 + an\sigma^2_{\beta}$
$(\tau\beta)_{ij}$	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta}$
$\varepsilon_{(ij)k}$	1	1	1	σ^2

E.g. for β_j: cover column j; rows containing j: β_j (remaining $a\cdot n$), $(\tau\beta)_{ij}$ (remaining $0\cdot n = 0$), ε $(1\cdot 1)$. So $\mathrm{E}(MS_B) = an\sigma^2_\beta + 0 + \sigma^2$ — the zero in the fixed column is exactly the restriction $\sum_i (\tau\beta)_{ij} = 0$. For τ_i: cover i; rows containing i: τ_i (bn), $(\tau\beta)_{ij}$ $(1\cdot n)$, ε (1).
With A random and B fixed the table is $\tau_i : [1, b, n]$, $\beta_j : [a, 0, n]$, $(\tau\beta)_{ij} : [1, 0, n]$, $\varepsilon : [1, 1, 1]$, df $a-1$, $b-1$, $(a-1)(b-1)$, $ab(n-1)$.

Example 9.14: Three factors, A and C random, B fixed

$y_{ijkl} = \mu + \tau_i + \beta_j + \gamma_k + (\tau\beta)_{ij} + (\tau\gamma)_{ik} + (\beta\gamma)_{jk} + (\tau\beta\gamma)_{ijk} + \varepsilon_{ijkl}$.

	R	F	R	R	
	a	b	c	n	
Term	i	j	k	l	EMS
τ_i	1	b	c	n	$\sigma^2 + bn\sigma^2_{\tau\gamma} + bcn\sigma^2_{\tau}$
β_j	a	0	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma} + acn\frac{\sum \beta_j^2}{b-1}$
γ_k	a	b	1	n	$\sigma^2 + bn\sigma^2_{\tau\gamma} + abn\sigma^2_{\gamma}$
$(\tau\beta)_{ij}$	1	0	c	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta}$
$(\tau\gamma)_{ik}$	1	b	1	n	$\sigma^2 + bn\sigma^2_{\tau\gamma}$
$(\beta\gamma)_{jk}$	a	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma} + an\sigma^2_{\beta\gamma}$
$(\tau\beta\gamma)_{ijk}$	1	0	1	n	$\sigma^2 + n\sigma^2_{\tau\beta\gamma}$
$\varepsilon_{(ijk)l}$	1	1	1	1	σ^2

Derivation for β_j: cover column j; the rows containing j are β_j $(a\cdot c\cdot n)$, $(\tau\beta)_{ij}$ $(1\cdot c\cdot n)$, $(\beta\gamma)_{jk}$ $(a\cdot 1\cdot n)$, $(\tau\beta\gamma)_{ijk}$ $(1\cdot 1\cdot n)$, ε (1). For τ_i: rows with i: τ_i (bcn), $(\tau\beta)_{ij}$ $(0\cdot cn = 0)$, $(\tau\gamma)_{ik}$ $(b\cdot 1\cdot n)$, $(\tau\beta\gamma)$ (0), ε.
Test for $H_0 : \beta_j = 0$. No single mean square has expectation $\sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma}$. But

$$\mathrm{E}(MS_{AB}) + \mathrm{E}(MS_{BC}) - \mathrm{E}(MS_{ABC}) = \sigma^2 + n\sigma^2_{\tau\beta\gamma} + cn\sigma^2_{\tau\beta} + an\sigma^2_{\beta\gamma},$$

so use the approximate (Satterthwaite) test

$$F_0 = \frac{MS_B}{MS_{AB} + MS_{BC} - MS_{ABC}} \quad \text{or} \quad F_0' = \frac{MS_B + MS_{ABC}}{MS_{AB} + MS_{BC}},$$

with denominator df $\nu = \dfrac{(MS_{AB} + MS_{BC} - MS_{ABC})^2}{\frac{MS_{AB}^2}{f_{AB}} + \frac{MS_{BC}^2}{f_{BC}} + \frac{MS_{ABC}^2}{f_{ABC}}}$ (and for F_0' both numerator and denominator df are of this form).